

Technology Stack

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Project Name: ShopEZ (E-Commerce Platform)

Components & Core Technologies

System Component	Purpose & Description	Chosen Technology
User Interface (Frontend)	Provides the visual layout, responsive grids, scalable vector icons, and dynamic theming engine for the user.	React.js 18, Vite, Vanilla CSS (CSS Variables), Lucide-React
Application Logic (Client)	Handles URL routing without page reloads, manages global application state, and orchestrates API network requests.	React Router v6, React Context API, Axios/Fetch API
Backend API (Server)	Exposes RESTful endpoints, processes complex business logic, and executes authentication middleware.	Node.js, Express.js 4.x, Bcryptjs, JsonWebToken (JWT)
Database (Persistence)	Provides flexible, document-oriented storage for complex entities like products, user profiles, and nested order records.	MongoDB, Mongoose 6.x ODM

Architectural Characteristics

Characteristic	Description	Implementation Technology
Security Implementations	Ensuring user data is protected against breaches and unauthorized access to administrative functions.	Bcryptjs for cryptographic hashing; JWT for tamper-proof, stateless authorization via HTTP Bearer tokens.
Scalable Architecture	Designing the system so that the frontend and backend can be hosted, scaled, and maintained completely independently.	Decoupled Express.js REST API serving data to a standalone React SPA.
Performance & Speed	Reducing initial load times and ensuring buttery smooth interactions once the application has loaded in the browser.	Vite bundler for highly optimized asset delivery; React Context API to minimize prop-drilling and redundant re-renders.